

ReactJS Hands-on - Consuming a REST API (fetchuserapp)

Fetching data from `api.randomuser.me` using the `fetch` API and `componentDidMount`

1. Objective

Create a React app **fetchuserapp** that retrieves user details from `https://api.randomuser.me/` and displays the title, first name and photo of a user, using a component named **Getuser** that calls the API inside the asynchronous `componentDidMount()` lifecycle method with `fetch()`.

2. Prerequisites

- Node.js
- NPM
- Visual Studio Code

3. Create the React App

● ● ● Command Prompt

```
C:\Users\student> npx create-react-app fetchuserapp
Creating a new React app in C:\Users\student\fetchuserapp.
Installing react, react-dom, and react-scripts...

Success! Created fetchuserapp at C:\Users\student\fetchuserapp
Happy hacking!

C:\Users\student> cd fetchuserapp
C:\Users\student\fetchuserapp> code .
```

4. Getuser Component

Added `Getuser.js` in the **src** folder. The component initialises state with an empty person object and a loading flag. In `componentDidMount()`, it awaits the `fetch()` call, parses the JSON response, and stores the first result in state; the render method then displays the title, first name and photo.

src/Getuser.js

```
import React, { Component } from 'react';

class Getuser extends Component {
  constructor(props) {
    super(props);
    this.state = {
      person: {},
      loading: true
    };
  }

  async componentDidMount() {
    const url = "https://api.randomuser.me/";
    const response = await fetch(url);
    const data = await response.json();
    this.setState({ person: data.results[0], loading: false });
    console.log(data.results[0]);
  }

  render() {
    const { person, loading } = this.state;

    if (loading) {
      return <p>Loading...</p>;
    }

    return (
      <div>
        <img src={person.picture.large} alt="user" />
        <h2>{person.name.title} {person.name.first}</h2>
      </div>
    );
  }
}

export default Getuser;
```

5. App.js

src/App.js

```
import React from 'react';
import Getuser from './Getuser';
```

```
function App() {  
  return (  
    <div className="App">  
      <Getuser />  
    </div>  
  );  
}  
  
export default App;
```

6. Run the Application

Command Prompt

```
C:\Users\student\fetchuserapp> npm start  
Compiled successfully!  
  
You can now view fetchuserapp in the browser.  
Local: http://localhost:3000
```

7. Output

On mount, the component fetches a random user and renders their photo, title and first name.

localhost:3000



Mr Donato

api.randomuser.me returns a different (random) person on every call, so the exact name and photo will differ each time the app is run - "Mr Donato" is the sample returned during this run.

8. Observation

componentDidMount() runs once, right after the component is first rendered to the DOM, which makes it the right place to kick off a network request - marking it async lets the fetch/await calls read naturally. Until the response arrives, state.loading stays true and the component shows a placeholder; once setState() runs with the fetched person, React re-renders and the actual title, name and photo appear.